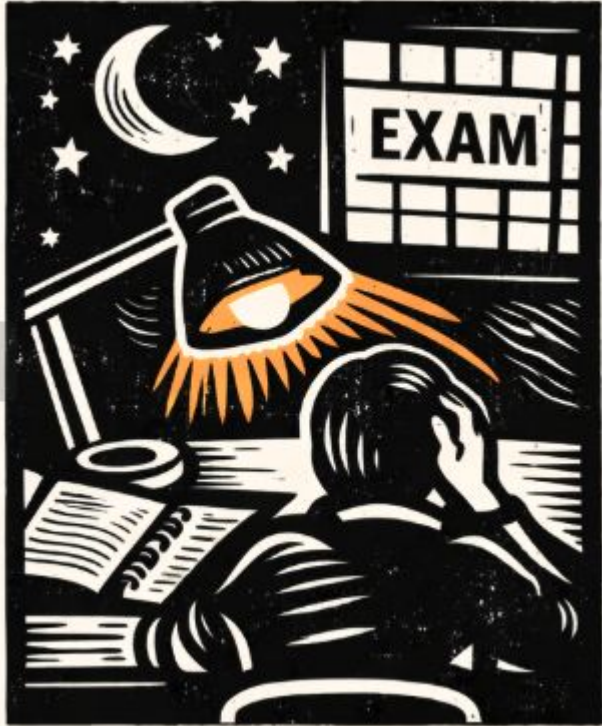


Enhancing Decision-Making with Data Analytics & Business Intelligence



Personal decision



You have an exam tomorrow.
You are trying to decide what
you should do tonight.

Managerial decision



You manage an online business. Yesterday's sales were below average. You're trying to decide how to respond to the data.

Strategic decision



You are one of the leaders of a growing company.
You're trying to decide if you should expand the business to another city.

Which of these is a semi-structured decision? Why?

- A. Deciding to change your major
- B. Responding to a drop in GPA
- C. Developing a personal budget
- D. Picking clothes for bad weather



Which of these is a semi-structured decision? Why?

- A. Deciding to change major:
(unstructured - long-term)
- B. Responding to a drop in GPA
(structured - clear rules)
- C. Developing a personal budget
(semi-structured - data + judgment)
- D. Picking clothes for bad weather
(structured - repeated + comparison)



From Decisions to Organizational Structure

- Operational
- Managerial
- Strategic



Where would you place the different decision types?

Unstructured,
structured, and
semi-structured
decisions

Operational
(bottom), middle
(center), and
senior (top)
management



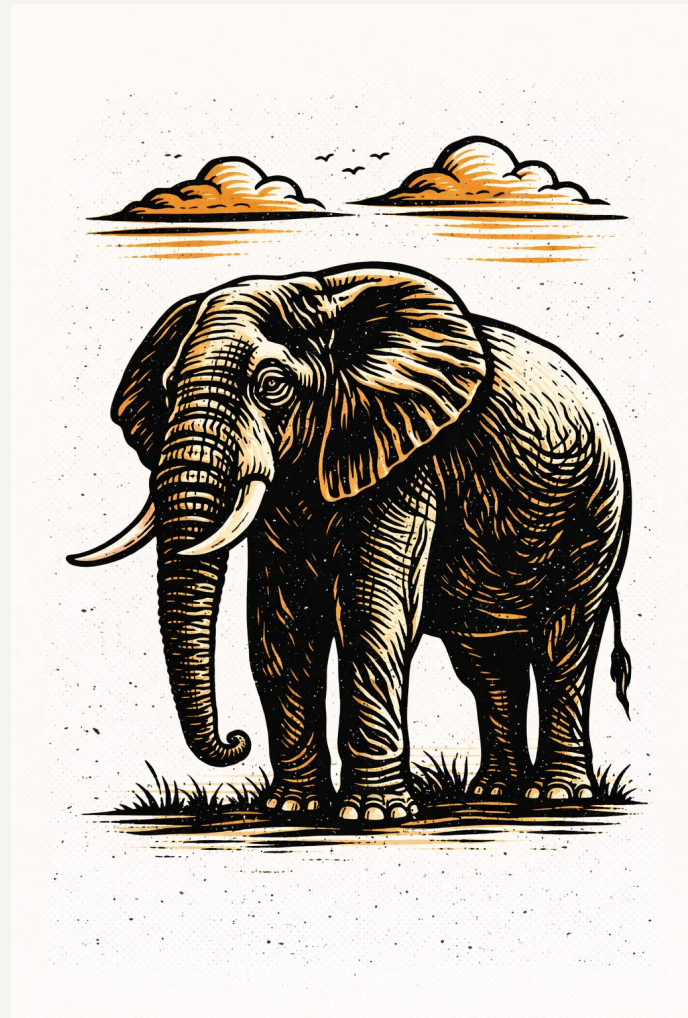
What kind of data are there?



- Structured
- Unstructured
- Semi-structured

What kind of data are there?

- Volume
- Variety
- Velocity
- Value
- Veracity

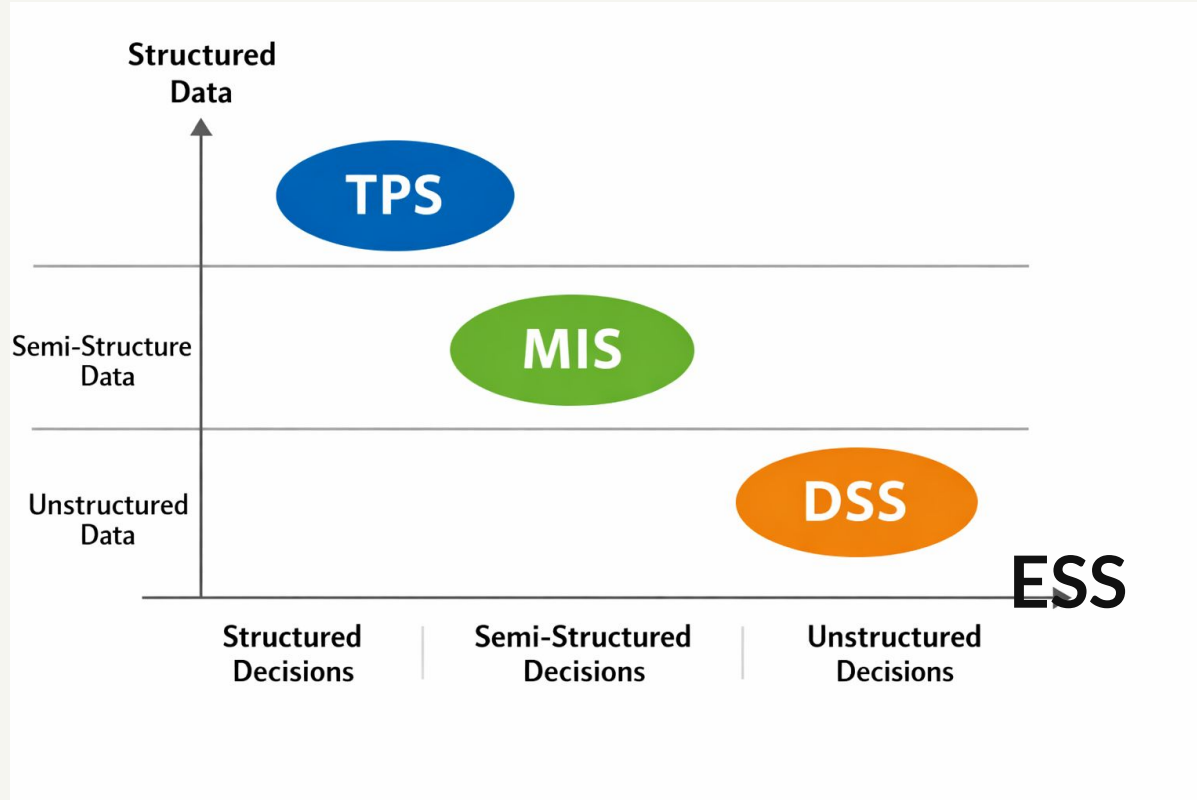


From Organization Structure to Systems

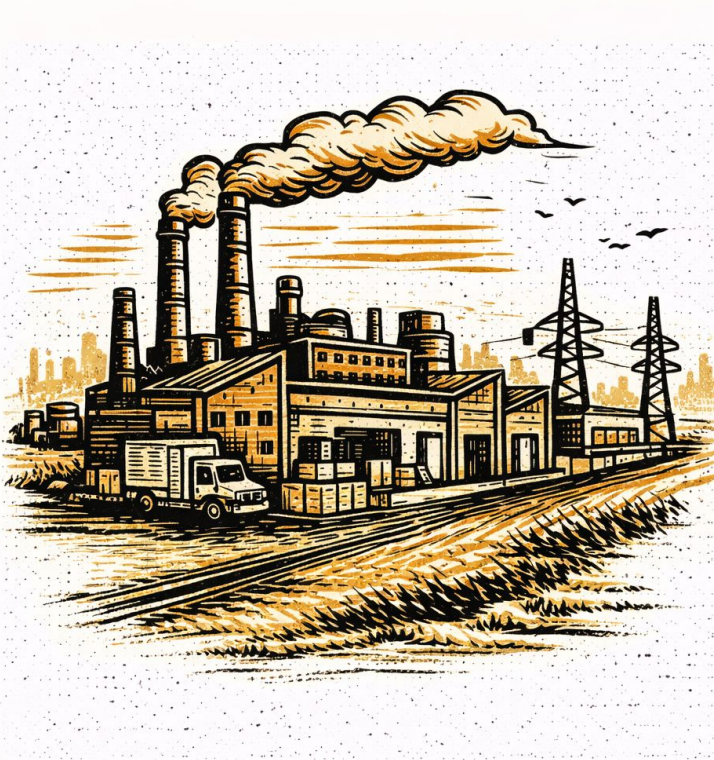
- Transaction Processing Systems (TPS)
- Management Information Systems (MIS)
- Decision Support Systems (DSS)
- Executive Support Systems (ESS)



Data & Decisions & Systems



From Organization Structure to Systems



- Customer Relationship Systems (CRM)
- Learning Management Systems (LMS)
- Supply-Chain Management Systems (SCM)
- Enterprise Resource Planning Systems (ERP)
- Office Automation Systems (OAS)

Do Information Systems Always Lead to Better Decisions?

- Data storage
- Data processing
- Structured information
- Operational control
- Standardization

- Data quality issues
- Information bias
- Management filters
- Organizational inertia
- Creativity blocker

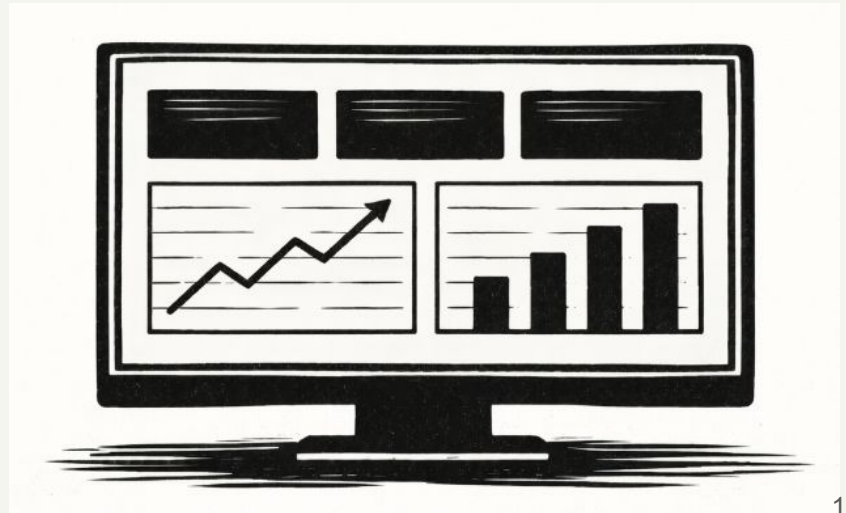
From Systems to Business Intelligence

- **Factory layer** =
TPS/MIS/ERP/CRM systems
- **Data layer** = Databases
and Data Warehouses
- **Intelligence layer** =
Dashboards and
Data Analytics



From BI Concepts to a Live Dashboard

Imagine you run a small online business.
Each morning you open your dashboard.
What information
helps you decide
what to do today?



CRM Sales Dashboard

Small online business · last 7 days

+ Add Order

Reset

Export CSV

Revenue

\$35,090.00

475 orders · 2.6%

+7.1% vs prior 7 days

Orders

475

18,260 sessions · AOV \$73.87

+6.0% vs prior 7 days

Conversion Rate

2.6%

2,490 add-to-carts · 850 checkouts

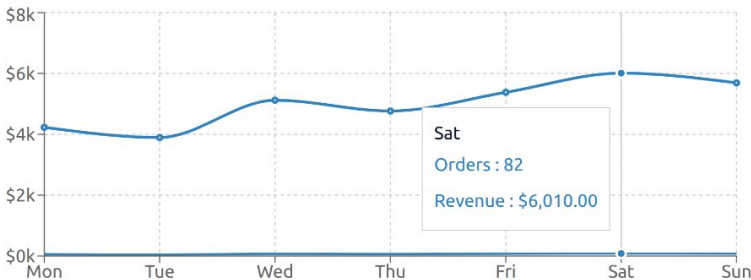
+2.1% vs prior 7 days

7-day Sales

Revenue trend (editable below)

Revenue

Orders



Recent Orders

Search and filter by channel/status

Search orders, customer, email, coupon...

All channels

All statuses

Showing 10 of 10 orders

Order	Customer	Channel	Total
LC-10479	Ethan Brooks ethan.brooks@example.com	Web	\$99.64
Mar 01, 06:05 PM	Delivered	US · Apple Pay	Subtotal \$92.00
	free-shipping	2 items	
LC-10478	Chloe Martin chloe.martin@example.com	Marketplace	\$46.50
Mar 01, 04:27 PM	Shipped	UK · Card	Subtotal \$38.00
	intl	1 item	

Summary: Enhancing decision-making

- **Data** input alone does not lead to better decisions.
- Information **systems** process data.
- Data **analysis** is one way of generating intelligence.
- Business **intelligence** output improves decisions.
- Generative AI can help build BI **dashboards**.

Outlook: What next?



- How is business success or failure measured?
- Which analytical models are most useful?
- Do information systems always improve decisions?
- What are the risks and benefits of using AI agents?

Online Lecture Materials

URL:

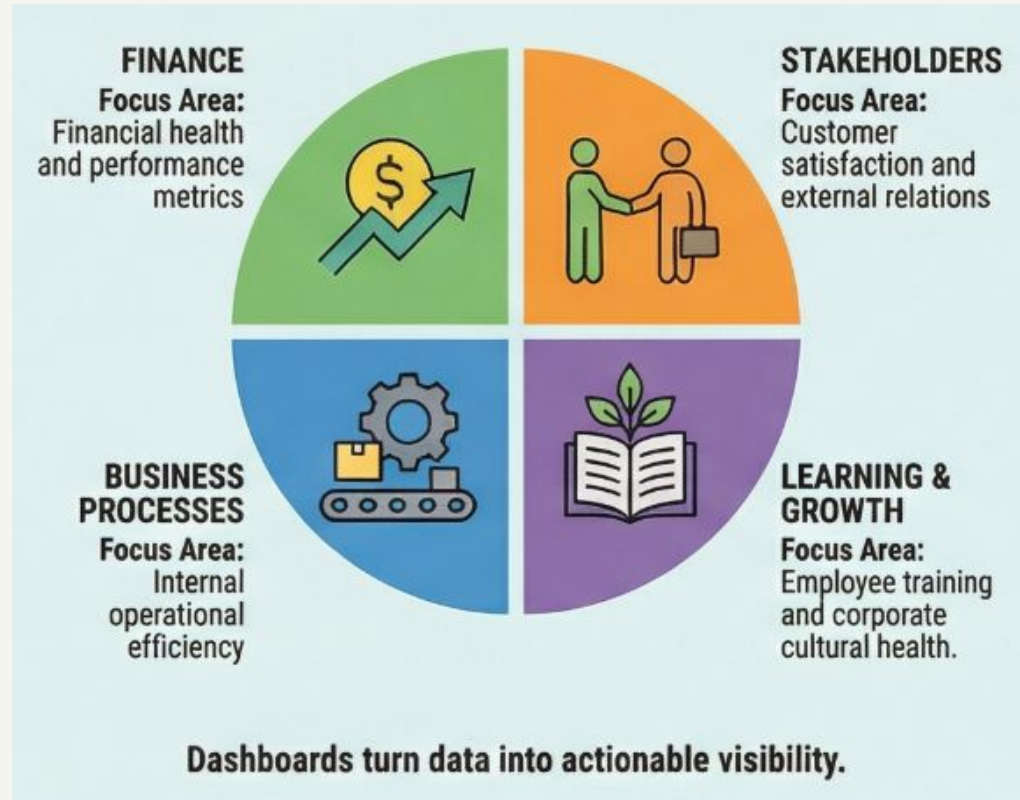
tinyurl.com/augustana-march-2026

- Lecture slides (PDF)
- Infographic overview
(generated by NotebookLM)
- Mock CRM sales dashboard
prompt (generated by ChatGPT)



Backup slides

Balanced Scorecard Framework



Designing a Dashboard: From Specification to System

Final refine: “I want to know whether yesterday’s sales were at least 10% below the average daily sales of last month, so that if they were, I will immediately launch a targeted email campaign to our existing customers today.”



Designing a Dashboard: From Specification to System

Now we have:

- Clear data: yesterday's sales; last month's average daily sales
- Explicit comparison: 10% below benchmark
- Concrete threshold: 10%
- Defined action: launch targeted email campaign
- Time frame: today

The specification is now precise enough
to build a system — and precise enough to act



Designing a dashboard: From Specification to System

1. Without data, you don't know what to do.
2. Without comparison, the number is meaningless.
3. Without a decision, the comparison is pointless.
4. Without specifics, the action cannot be taken.



Designing a dashboard: From Specification to Prompt

Create an interactive dashboard component (not a chat response).

Render it as a live, editable UI panel in the side preview.

Use sample CRM sales data for a small online business.

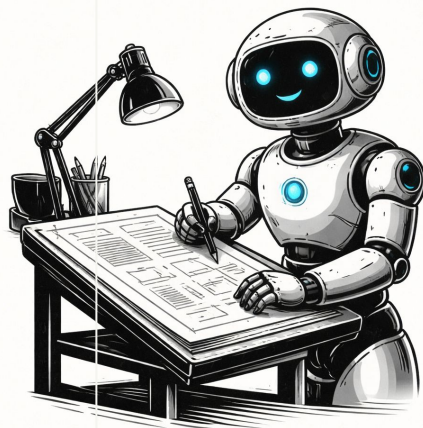
Include KPI cards (Revenue, Orders, Conversion Rate),
a 7-day sales chart, and a recent orders table.

Pre-populate with realistic data.

Do not explain anything.

Do not output markdown or prose.

Generate only the interactive dashboard preview.



Designing a Dashboard: Result

